VIDYAVARDHAKA COLLEGE OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

ABA Activity Report:

Production Cost Analysis of a Manufactured Part

Course Name	Management and Engineering Economics
Course Code	BMEMA503
Course Instructor	Dr. Nithyananda B S, Associate Professor
Department	Mechanical Engineering
Academic Year	2025-26 (Odd Semester)
Sem and Section	5 'A'

Report Prepared by

Student Name	USN	Signature
BHAVISI GANGAMMA A A	4VV23ME011	
BHOOMIKA K N	4VV23ME012	
HARINI S	4VV23ME031	
HARSHITHA S	4VV23ME035	

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COMPANY PROFILE

Inchametric Gears Ltd in mysore Inchametric Gears Ltd is a leading B2B company located in mysore, known for its expertise and innovative approach in the industry. With a profound understanding and a wealth of experience, the company transforms complex challenges into innovative solutions, tailored to meet the unique needs of businesses across various sectors like Gear Manufacturers. The company has established a strong reputation for excellence by delivering products and services that drive growth and innovation in the market. Every project or partnership undertaken by Inchametric Gears Ltd demonstrates a commitment to quality, efficiency, and innovation. The company's success is backed by a dynamic team of seasoned professionals, including engineers, consultants, project managers, product specialists, customer service experts, and research and development professionals, all of whom are adept at providing customized solutions aligned with emerging industry trends.

The company follows a meticulous process of consultation, planning, execution, and delivery for every engagement. Inchametric Gears Ltd offers a wide range of products and services, including but not limited to customized manufacturing solutions, management, retail, and logistics solutions.

SELECTED COMPONENT DETAILS

We selected spur gears as our component of study at Inchametric Gears Ltd because they are one of the primary power transmission products manufactured in the company. Spur gears are the simplest and most widely used gear type, featuring straight teeth and transmitting motion between parallel shafts.

In addition to spur gears, Inchametric Gears Ltd also manufactures helical gears and various other power transmission components, showcasing their broad capability in gear manufacturing . The company produces spur gears using high-quality materials such as alloy steel and stainless steel, ensuring strength, durability, and performance. All gears are manufactured according to international standards such as DIN, AGMA, BS, and IS, guaranteeing accuracy and quality.

Their production capability ranges from 20 mm to 500 mm in diameter, and they use advanced technologies like CNC machining, hobbing, shaping, and profile grinding. Inchametric can also produce ground spur gears up to DIN Class 3–4, demonstrating high precision and smooth operation. Because spur gears are widely used in industrial and automotive applications, and because Inchametric excels in producing them with world-class accuracy, they were chosen as our selected component for study.

COST COMPONENT ANALYSIS

SL.NO	PROCESS/ITEM	COST
1	Raw Material (Alloy steel , Forging, approx. 1.2 Kg)	220
2	Initial Machining and Cutting	50
3	Gear Hobbing / Cutting (CNC)	180
4	Drilling / Boring (1 Hole)	20
5	Heat Treatment (Hardening)	120
6	Precision Grinding (ID/OD/Profile)	150
7	Additional Charges (Inspection , Consumables)	60
8	Factory Overhead (Indirect , Labour, Electricity, etc)	300
9	Administrative Overhead (Office, Marketing,etc)	150
10	TOTAL COST	1250
	SELLING PRICE (Approximate Market Rate)	1800

KEY LEARNINGS

Customization is Key: Inchametric Gears Ltd focuses on providing customized gear solutions to meet specific client needs across diverse industries like automotive and hydraulics, rather than just selling standard off-the-shelf items.

Versatile Manufacturing Capabilities: The company can produce a wide range of gear types beyond just spur gears, including helical gears, transmission shafts, crankshafts, and internal gears, highlighting a comprehensive engineering ability.

Dual Standard Expertise: The very name "Inchametric" indicates expertise in manufacturing products to both inch and metric specifications and standards (e.g., AGMA, DIN), which is a key advantage for serving a global market.

Precision Engineering: High-quality manufacturing involves advanced processes such as CNC machining, hobbing, shaping, and precision grinding to achieve strict quality standards, with reported capabilities up to DIN Class 3-4 for hardened gears**.**

Material Selection Matters: The choice of material (alloy steel, stainless steel, etc.) is crucial and depends on the specific application's requirements for strength, durability, and operating conditions**.**

Cost Drivers are Complex: The final cost of a gear is influenced by proprietary factors like material type, size, precision requirements, and batch quantity. Publicly available information only provides general market price ranges, not internal cost breakdowns (material, labor, overheads).

Regional Hub: Mysore, specifically industrial areas like Metagalli and Hootagalli, is home to several established and experienced gear manufacturers, indicating a strong local engineering ecosystem in Karnataka, India.

REFERENCES

The information in this report was collected through direct interactions with the labourers, machine operators, and accounts staff of the industry, who provided practical insights into the manufacturing processes and internal operations.

RUBICS OF EVALUATION:

Criteria	Description	Marks
A. Understanding of Company Profile and Process	Clarity in presenting the company background, manufacturing process flow, and relevance of the selected part.	1
B. Identification of the Part and Its Specifications	Proper selection of part, drawing/sketch/photo, material details, and understanding of function.	2
C. Cost Component Analysis	Correct identification and calculation/estimation of cost elements such as raw material, machining, labor, overheads, etc.	3
D. Report Quality & Presentation	Clarity, organization, data representation (charts/tables/photos), formatting, and conclusions.	2
E. Teamwork & Industrial Interaction	Evidence of teamwork, interaction with industry personnel, and distribution of work among members.	2

PO	Program Outcome Description	Relevance to Activity	Mapping Level
PO2	*Problem Analysis:* Identify, formulate, and analyze complex engineering problems considering sustainability.	Students analyze cost-driving factors and propose feasible cost optimization measures.	3
PO8	Individual and Collaborative Team Work: Function effectively as an individual and as a member or leader in a team.	Students work in teams during industrial visits and collaboratively compile the report.	3
PO9	Communication: Communicate effectively and inclusively within the engineering community and society.	Students prepare and present professional technical reports and visual summaries.	3
PO10	Project Management and Finance: Apply principles of engineering management and economics in project contexts.	Students apply cost and finance principles to real manufacturing scenarios.	3

SDG No.	Goal	Relevance to the Activity
SDG 4	Quality Education	Promotes experiential learning through industry-academia linkage.
SDG 8	Decent Work and Economic Growth	Enhances understanding of industrial efficiency and productivity economics.
SDG 9	Industry, Innovation, and Infrastructure	Encourages engagement with real manufacturing processes and innovation.
SDG 12	Responsible Consumption and Production	Fosters awareness of sustainable and cost-effective manufacturing practices.

Evaluation Table

Student Name	USN	A	B	C	D	E	Total Marks
BHAVISI GANGAMMA A A	4VV23ME011						
BHOOMIKA K N	4VV23ME012						
HARINI S	4VV23ME031						
HARSHITHA S	4VV23ME035						

Signature of Course Instructor